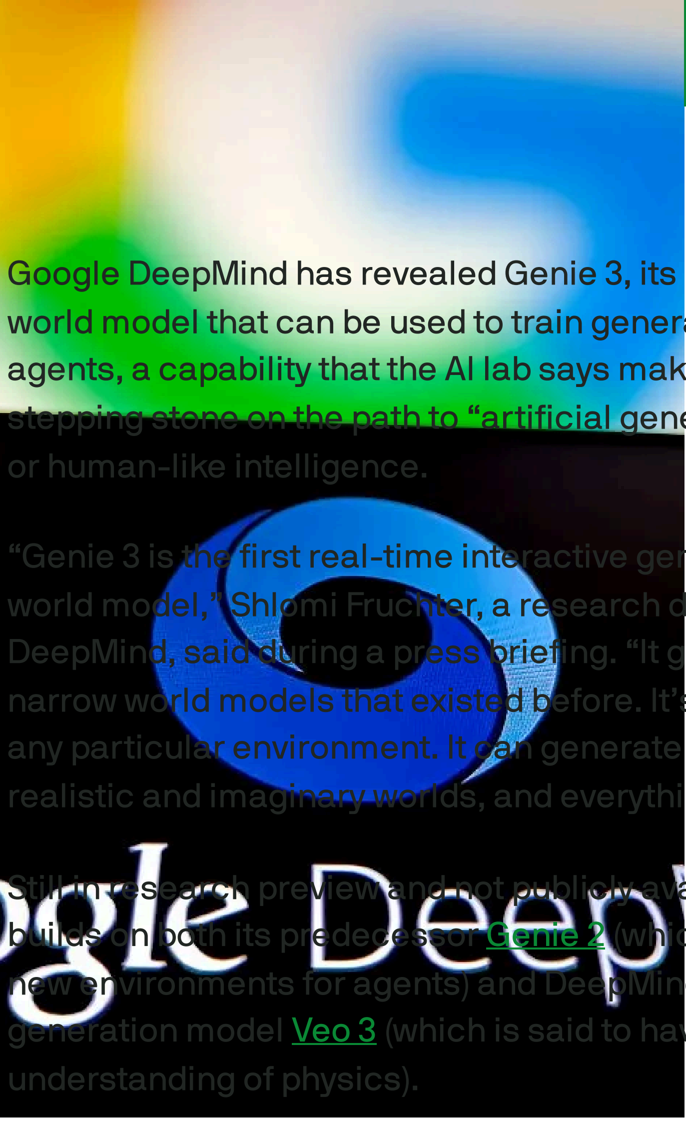
AI

# DeepMind thinks its new Genie 3 world model presents a stepping stone toward AGI

Rebecca Bellan — 7:10 AM PDT · August 5, 2025



Google DeepMind has revealed Genie 3, its latest foundation world model that can be used to train general-purpose AI agents, a capability that the AI lab says makes for a crucial stepping stone on the path to "artificial general intelligence," or human-like intelligence.

"Genie 3 is the first real-time interactive general-purpose world model," Shlomi Fruchter, a research director at DeepMind, said during a press briefing. "It goes beyond narrow world models that existed before. It's not specific to any particular environment. It can generate both photo-realistic and imaginary worlds, and everything in between."

Still in research preview and not publicly available, Genie 3 builds on both its predecessor Genie 2 (which can generate new environments for agents) and DeepMind's latest video generation model Veo 3 (which is said to have a deep understanding of physics).

IMAGE CREDITS: VCG/VCG / GETTY IMAGES

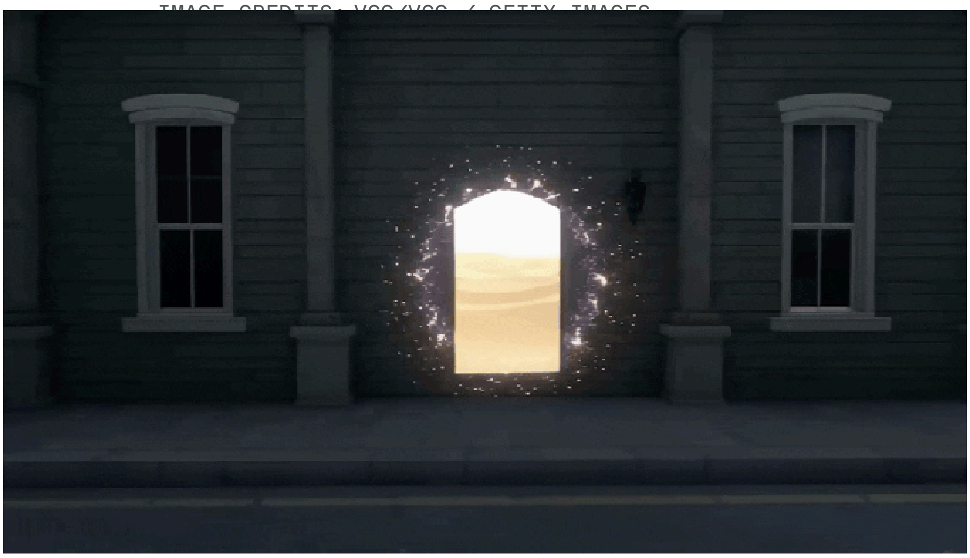


IMAGE CREDITS: GOOGLE DEEPMIND

With a simple text prompt, Genie 3 can generate multiple minutes of interactive 3D environments at 720p resolution at 24 frames per second — a significant jump from the 10 to 20 seconds Genie 2 could produce. The model also features "promptable world events," or the ability to use a prompt to change the generated world.

Perhaps most importantly, Genie 3's simulations stay physically consistent over time because the model can

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remember what it previously generated — a capability that DeepMind says its researchers didn't explicitly program into the model.

Fruchter said that while Genie 3 has implications for educational experiences, gaming or prototyping creative concepts, its real unlock will manifest in training agents for general-purpose tasks, which he said is essential to reaching AGI.

"We think world models are key on the path to AGI, specifically for embodied agents, where simulating real world scenarios is particularly challenging," Jack Parker-Holder, a research scientist on DeepMind's open-endedness team, said during the briefing.



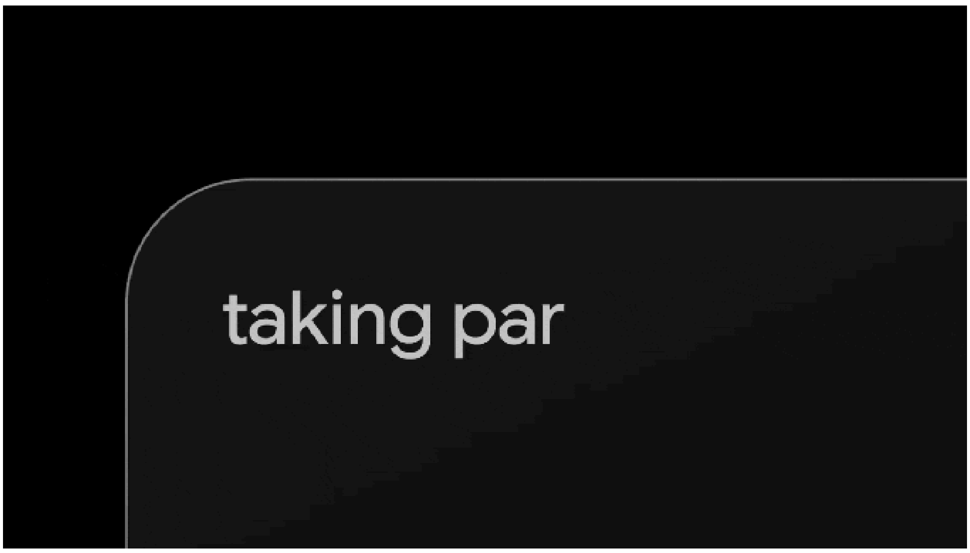


IMAGE CREDITS:GOOGLE DEEPMIND

Genie 3 is supposedly designed to solve that bottleneck. Like Veo, it doesn't rely on a hard-coded physics engine; instead, DeepMind says, the model teaches itself how the world works — how objects move, fall, and interact — by remembering what it has generated and reasoning over long time horizons.

"The model is auto-regressive, meaning it generates one frame at a time," Fruchter told TechCrunch in an interview. "It has to look back at what was generated before to decide what's going to happen next. That's a key part of the architecture."

That memory, the company says, lends to consistency in Genie 3's simulated worlds, which in turn allows it to develop a grasp of physics, similar to how humans understand that a glass teetering on the edge of a table is about to fall, or that they should duck to avoid a falling object.

Notably, DeepMind says the model also has the potential to push AI agents to their limits — forcing them to learn from their own experience, similar to how humans learn in the real world.

As an example, DeepMind shared its test of Genie 3 with a recent version of its generalist Scalable Instructable Multiworld Agent (SIMA), instructing it to pursue a set of goals. In a warehouse setting, they asked the agent to perform tasks like "approach the bright green trash compactor" or "walk to the packed red forklift."

"In all three cases, the SIMA agent is able to achieve the goal," Parker-Holder said. "It just receives the actions from the agent. So the agent takes the goal, sees the world simulated around it, and then takes the actions in the world. Genie 3 simulates forward, and the fact that it's able to achieve it is because Genie 3 remains consistent."

IMAGE CREDITS:GOOGLE DEEPMIND

That said, Genie 3 has its limitations. For example, while the researchers claim it can understand physics, the demo showing a skier barreling down a mountain didn't reflect how snow would move in relation to the skier.

Additionally, the range of actions an agent can take is limited. For example, the promptable world events allow for a wide range of environmental interventions, but they're not necessarily performed by the agent itself. And it's still difficult to accurately model complex interactions between multiple independent agents in a shared environment.

Genie 3 can also only support a few minutes of continuous interaction, when hours would be necessary for proper training.

Still, the model presents a compelling step forward in teaching agents to go beyond reacting to inputs, letting them potentially plan, explore, seek out uncertainty, and improve through trial and error — the kind of self-driven, embodied learning that many say is key to moving toward general intelligence.

"We haven't really had a Move 37 moment for embodied agents yet, where they can actually take novel actions in the real world," Parker-Holder said, referring to the legendary moment in the 2016 game of Go between DeepMind's AI agent AlphaGo and world champion Lee Sedol, in which Alpha Go played an unconventional and brilliant move that became symbolic of AI's ability to discover new strategies beyond human understanding.

"But now, we can potentially usher in a new era," he said.

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Rebecca Bellan is a senior reporter at TechCrunch, where she covers Tesla and Elon Musk's broader empire, autonomy, AI, electrification, g...

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AI

Google is forming a new team to build AI that can simulate the physical world

Kyle Wiggers 10:56 AM PST · January 6, 2025

Google is forming a new team to work on AI models that can simulate the physical world.

Tim Brooks — one of the co-leads on OpenAI's video generator, Sora, who left for Google's AI research lab, Google DeepMind, in October — will lead the new team, he announced in a post on X. It'll be a part of Google DeepMind.

"DeepMind has ambitious plans to make massive generative models that simulate the world," Brooks wrote Monday morning. "I'm hiring for a new team with this mission."

According to job listings Brooks linked to in his post, the new modeling team will collaborate with and build on work from Google's Gemini, Veo, and Genie teams to tackle "critical new problems" and scale models "to the highest levels of compute." Gemini is Google's flagship series of AI models for tasks like analyzing images and generating text, while Veo is Google's own video generation model.

As for Genie, it's Google's take on a world model — AI that can simulate games and 3D environments in real time. Google's latest Genie model, previewed in December, can generate a massive variety of playable 3D worlds.

existing multimodal models such as Gemini.



A number of startups and big tech companies are chasing after world models, including influential AI researcher Fei-Fei Lee's World Labs, Israeli upstart Decart, and Odyssey. They believe that world models could one day be used to create interactive media, like video games and movies, and run realistic simulations like training environments for robots.

> Come work with Tim and the Deepmind team on massive world simulation models : )
>
> On the critical path to AGI. https://t.co/4Zuju5eMHb
>
> — Logan Kilpatrick (@OfficialLoganK) January 6, 2025

But creatives have mixed feelings about the tech.

A recent Wired investigation found that game studios like Activision Blizzard, which has laid off scores of workers, are using AI to cut corners, ramp up productivity, and compensate for attrition. And a 2024 study commissioned by the Animation Guild, a union representing Hollywood animators and cartoonists, estimated that over 100,000 U.S.-based film, television, and animation jobs will be disrupted by AI by 2026.

Some startups in the nascent world modeling space, like Odyssey, have pledged to collaborate with creative professionals — not replace them. We'll have to see if Google follows suit.

There's also the unresolved matter of copyright. Some world models appear to be trained on clips of video game playthroughs, which could make the companies developing those models the target of lawsuits in cases where the videos were unlicensed.

Google, which owns YouTube, asserts that it has permission to train its models on YouTube videos in accordance with the platform's terms of service. But the company hasn't said which specific videos it is sourcing for training.

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**Kyle Wiggers**
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Kyle Wiggers was TechCrunch's AI Editor until June 2025. His writing has appeared in VentureBeat and Digital Trends, as well as a range of...

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# DeepMind claims its AI performs better than International Mathematical Olympiad gold medalists

Kyle Wiggers — 10:56 AM PST · February 7, 2025

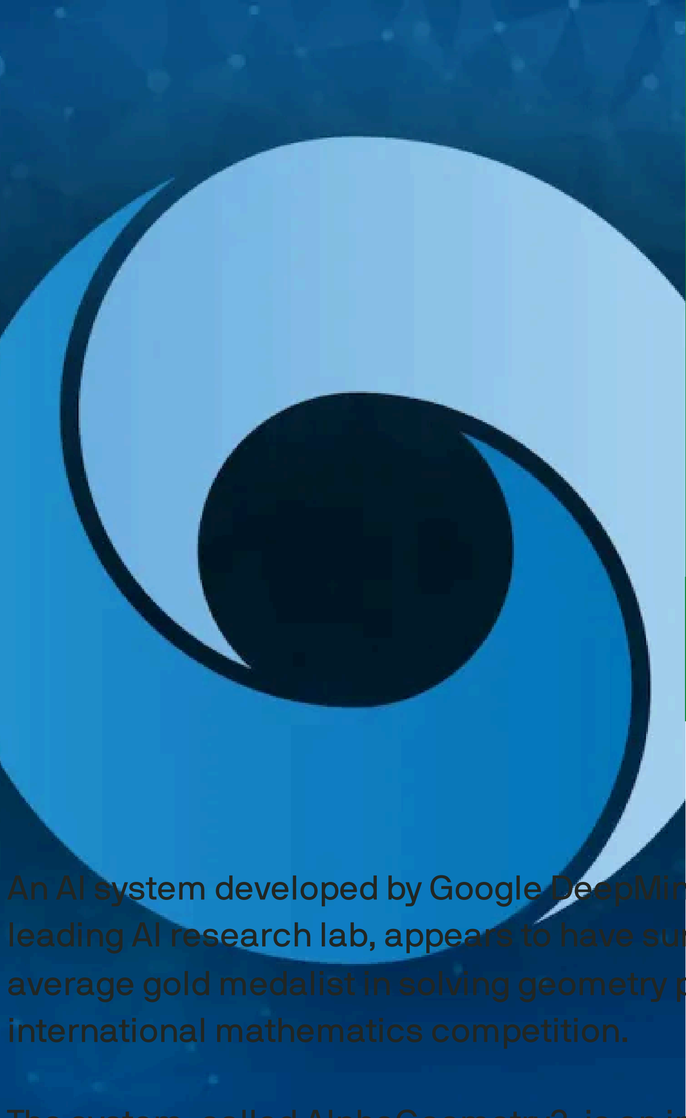

IMAGE CREDITS: DEEPMIND

An AI system developed by Google DeepMind, Google's leading AI research lab, appears to have surpassed the average gold medalist in solving geometry problems in an international mathematics competition.

The system, called AlphaGeometry2, is an improved version of a system, AlphaGeometry, [that DeepMind released last January](). In a [newly published study](), the DeepMind researchers behind AlphaGeometry2 claim their AI can solve 84% of all geometry problems over the last 25 years in the International Mathematical Olympiad (IMO), a math contest for high school students.

Why does DeepMind care about a high-school-level math competition? Well, the lab thinks the key to more capable AI might lie in discovering new ways to solve challenging geometry problems — specifically [Euclidean geometry problems]().

Proving mathematical theorems, or logically explaining why a theorem (e.g. the Pythagorean theorem) is true, requires both reasoning and the ability to choose from a range of possible steps toward a solution. These problem-solving skills could — if DeepMind's right — turn out to be a useful component of future general-purpose AI models.

Indeed, this past summer, DeepMind demoed a system that combined AlphaGeometry2 with AlphaProof, an AI model for formal math reasoning, to solve four out of six problems from



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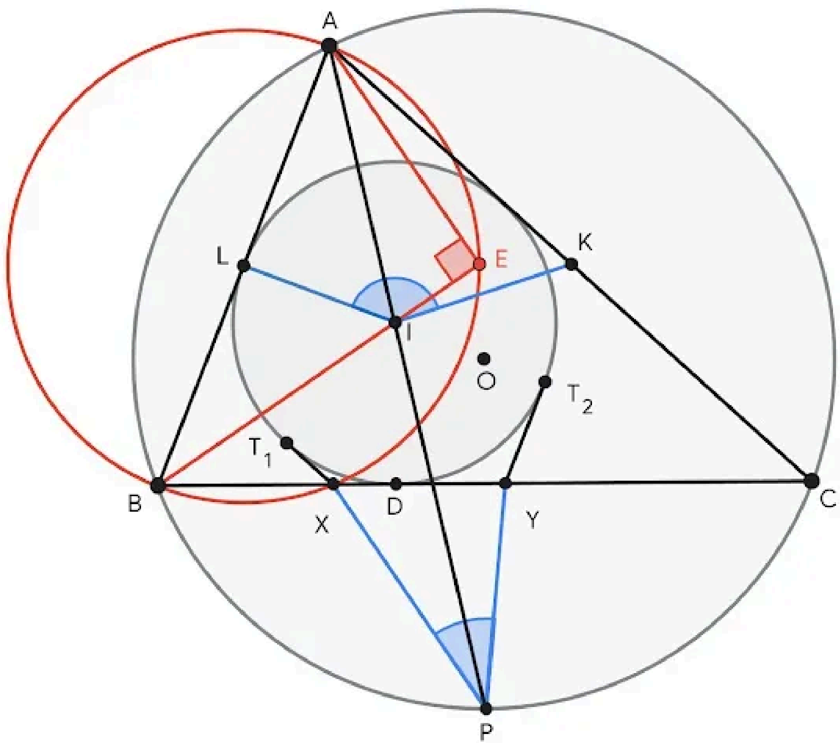
the 2024 IMO. In addition to geometry problems, approaches like these could be extended to other areas of math and science — for example, to aid with complex engineering calculations.

AlphaGeometry2 has several core elements, including a language model from Google's Gemini family of AI models and a "symbolic engine." The Gemini model helps the symbolic engine, which uses mathematical rules to infer solutions to problems, arrive at feasible proofs for a given geometry theorem.




A TYPICAL GEOMETRY PROBLEM DIAGRAM IN AN IMO EXAM. IMAGE CREDITS:GOOGLE

Olympiad geometry problems are based on diagrams that need "constructs" to be added before they can be solved, such as points, lines, or circles. AlphaGeometry2's Gemini model predicts which constructs might be useful to add to a diagram, which the engine references to make deductions.

Basically, AlphaGeometry2's Gemini model suggests steps and constructions in a formal mathematical language to the engine, which — following specific rules — checks these steps for logical consistency. A search algorithm allows AlphaGeometry2 to conduct multiple searches for solutions in parallel and store possibly useful findings in a common knowledge base.

AlphaGeometry2 considers a problem to be "solved" when it arrives at a proof that combines the Gemini model's suggestions with the symbolic engine's known principles.

Owing to the complexities of translating proofs into a format AI can understand, there's a dearth of usable geometry training data. So DeepMind created its own synthetic data to train AlphaGeometry2's language model, generating over 300 million theorems and proofs of varying complexity.

The DeepMind team selected 45 geometry problems from IMO competitions over the past 25 years (from 2000 to 2024), including linear equations and equations that require moving geometric objects around a plane. They then "translated" these into a larger set of 50 problems. (For technical reasons, some problems had to be split into two.)

According to the paper, AlphaGeometry2 solved 42 out of the 50 problems, clearing the average gold medalist score of 40.9.

Granted, there are limitations. A technical quirk prevents AlphaGeometry2 from solving problems with a variable number of points, nonlinear equations, and inequalities. And AlphaGeometry2 isn't *technically* the first AI system to reach gold-medal-level performance in geometry, although it's the first to achieve it with a problem set of this size.

AlphaGeometry2 also did worse on another set of harder IMO problems. For an added challenge, the DeepMind team selected problems — 29 in total — that had been nominated for IMO exams by math experts, but that haven't yet appeared in a competition. AlphaGeometry2 could only solve 20 of these.

Still, the study results are likely to fuel the debate over whether AI systems should be built on symbol manipulation — that is, manipulating symbols that represent knowledge using rules — or the ostensibly more brain-like neural networks.

AlphaGeometry2 adopts a hybrid approach: Its Gemini model has a neural network architecture, while its symbolic engine is rules-based.

Proponents of neural network techniques argue that intelligent behavior, from speech recognition to image generation, can emerge from nothing more than massive amounts of data and computing. Opposed to symbolic systems, which solve tasks by defining sets of symbol-manipulating rules dedicated to particular jobs, like editing a line in word processor software, neural networks try to solve tasks through statistical approximation and learning from examples.

Neural networks are the cornerstone of powerful AI systems like OpenAI's o1 "reasoning" model. But, claim supporters of symbolic AI, they're not the end-all-be-all; symbolic AI might be better positioned to efficiently encode the world's knowledge, reason their way through complex scenarios, and "explain" how they arrived at an answer, these supporters argue.

"It is striking to see the contrast between continuing, spectacular progress on these kinds of benchmarks, and meanwhile, language models, including more recent ones with 'reasoning,' continuing to struggle with some simple commonsense problems," Vince Conitzer, a Carnegie Mellon University computer science professor specializing in AI, told TechCrunch. "I don't think it's all smoke and mirrors, but it illustrates that we still don't really know what behavior to expect from the next system. These systems are likely to be very impactful, so we urgently need to understand them and the risks they pose much better."

AlphaGeometry2 perhaps demonstrates that the two approaches — symbol manipulation and neural networks — *combined* are a promising path forward in the search for generalizable AI. Indeed, according to the DeepMind paper, o1, which also has a neural network architecture, couldn't solve any of the IMO problems that AlphaGeometry2 was able to answer.

This may not be the case forever. In the paper, the DeepMind team said it found preliminary evidence that AlphaGeometry2's language model was capable of generating partial solutions to problems without the help of the symbolic engine.

"[The] results support ideas that large language models can be self-sufficient without depending on external tools [like symbolic engines]," the DeepMind team wrote in the paper, "but until [model] speed is improved and hallucinations are completely resolved, the tools will stay essential for math applications."

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**Kyle Wiggers**

AI Editor |

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